

TAMPERE HARMALA, Finland

WMO: 027630

Lat: 61.4656N

Lon: 23.7468E

Elev: 86

StdP: 100.30

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.2	(c) -20.2	(d) -25.8	(e) 0.4	(f) -23.2	(g) -22.5	(h) 0.5	(i) -19.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	27.6	18.6	25.5	17.4	23.6	16.6	20.2	24.5	18.9	23.5	17.7	22.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	22.2	17.2	12.4	20.8	16.0	11.5	19.5	58.2	24.5	54.1	23.5	50.2	22.2	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -24.6	(f) 29.0	(g) 4.3	(h) 2.4	(i) -27.7	(j) 30.7	(k) -30.2	(l) 32.1	(m) -32.6	(n) 33.4	(o) -35.8	(p) 35.2		
(4) N/A	(5) N/A	(6) N/A	(e) -24.8	(f) 20.9	(g) 4.3	(h) 1.9	(i) -27.9	(j) 22.3	(k) -30.4	(l) 23.4	(m) -32.8	(n) 24.4	(o) -36.0	(p) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-6.9	-4.9	-1.9	4.0	10.5	14.1	17.4	15.5	10.9	4.6	1.2	-2.8
	DBStd	9.18	6.44	6.31	4.66	3.50	4.20	3.46	3.25	2.95	3.23	3.58	4.23	5.59
	HDD10.0	2403	525	418	368	184	45	5	0	1	28	169	264	396
	HDD18.3	4842	783	652	627	431	245	134	58	96	223	425	514	654
	CDD10.0	649	0	0	0	3	60	128	228	172	55	3	0	0
	CDD18.3	46	0	0	0	0	2	7	28	9	0	0	0	0
	CDH23.3	468	0	0	0	0	39	93	271	65	0	0	0	0
	CDH26.7	88	0	0	0	0	3	17	57	11	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	566	36	26	29	33	36	51	71	74	60	55	53	42
	PrecMax	863	74	65	60	89	89	169	132	183	112	129	97	91
	PrecMin	402	12	4	1	1	7	16	21	22	6	12	10	7
	PrecStd	97	16	15	16	20	17	33	34	38	25	29	23	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.8	10.5	19.9	26.7	28.9	30.3	28.4	21.5	14.1	10.3	8.4
		MCWB	3.5	3.2	5.1	10.5	15.3	18.3	19.6	19.7	16.5	11.8	9.3	7.8
	2%	DB	2.7	3.7	7.3	16.1	24.2	26.1	28.4	25.0	19.2	12.2	8.9	5.4
		MCWB	2.1	2.5	3.7	8.5	14.4	16.7	19.4	17.5	15.0	10.6	8.3	4.6
	5%	DB	1.7	2.7	5.4	13.3	21.8	23.6	26.4	23.0	17.5	11.3	7.4	4.1
		MCWB	1.1	2.0	2.7	7.2	13.6	15.8	18.5	16.6	14.0	10.0	6.8	3.5
	10%	DB	0.8	1.7	3.8	10.7	19.3	21.5	24.4	21.3	16.1	10.2	6.2	3.0
		MCWB	0.4	1.1	1.8	5.9	12.2	14.9	17.9	15.9	13.6	9.0	5.7	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.6	6.0	11.5	17.2	20.2	21.8	21.0	17.0	12.5	9.5	8.0
		MCDB	3.9	4.6	8.7	17.5	22.5	25.2	27.3	25.4	20.2	13.8	10.1	8.3
	2%	WB	2.3	2.8	4.2	9.3	15.7	18.1	20.7	19.1	15.9	11.3	8.3	4.7
		MCDB	2.7	3.4	6.6	14.8	22.3	23.7	25.5	22.9	18.2	12.0	9.0	5.2
	5%	WB	1.2	2.0	3.1	7.7	14.4	16.8	19.7	17.9	14.9	10.3	6.9	3.5
		MCDB	1.5	2.6	4.8	11.9	20.2	22.2	24.0	21.3	16.8	11.0	7.3	3.9
	10%	WB	0.4	1.2	2.2	6.4	12.8	15.6	18.6	16.8	13.9	9.0	5.8	2.5
		MCDB	0.8	1.7	3.5	10.2	18.3	20.2	23.3	20.0	15.5	10.1	6.2	2.9
Mean Daily Temperature Range	5% DB	MDBR	5.8	5.5	7.8	9.6	10.9	9.8	9.6	9.2	7.7	6.0	3.8	4.3
		MCDBR	3.5	5.0	9.0	15.1	16.0	14.1	12.6	12.0	10.1	7.1	4.4	4.4
		MCWBR	3.5	4.2	6.0	8.4	7.3	6.5	5.2	5.9	6.3	5.5	4.0	4.2
	5% WB	MCDBR	3.5	4.6	8.1	12.7	13.4	11.6	10.7	9.9	8.5	6.0	4.2	4.4
		MCWBR	3.5	4.1	5.8	7.6	6.8	5.9	5.2	5.3	5.6	5.4	4.0	4.2
Clear-Sky Solar Irradiance	taub	0.251	0.295	0.322	0.356	0.353	0.349	0.374	0.360	0.336	0.321	0.274	0.357	
	taud	2.085	2.166	2.252	2.372	2.444	2.477	2.426	2.456	2.519	2.411	2.184	2.597	
	Ebn at Noon	452	665	788	836	870	879	843	825	785	648	423	300	
	Edh at Noon	38	69	90	98	99	98	100	90	71	56	35	24	
All-Sky Solar Radiation	RadAvg	0.22	0.78	2.25	3.68	5.01	5.32	5.03	3.94	2.34	0.99	0.26	0.12	
	RadStd	0.04	0.18	0.28	0.42	0.54	0.41	0.51	0.44	0.30	0.19	0.05	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	-134	N/A	N/A	N/A

Nomenclature: See separate page